Lab experiment-25

Experiment 25. Prolog Program for Fruit and Its Color Using Backtracking

Aim

To write a Prolog program to find the **color of fruits using backtracking**.

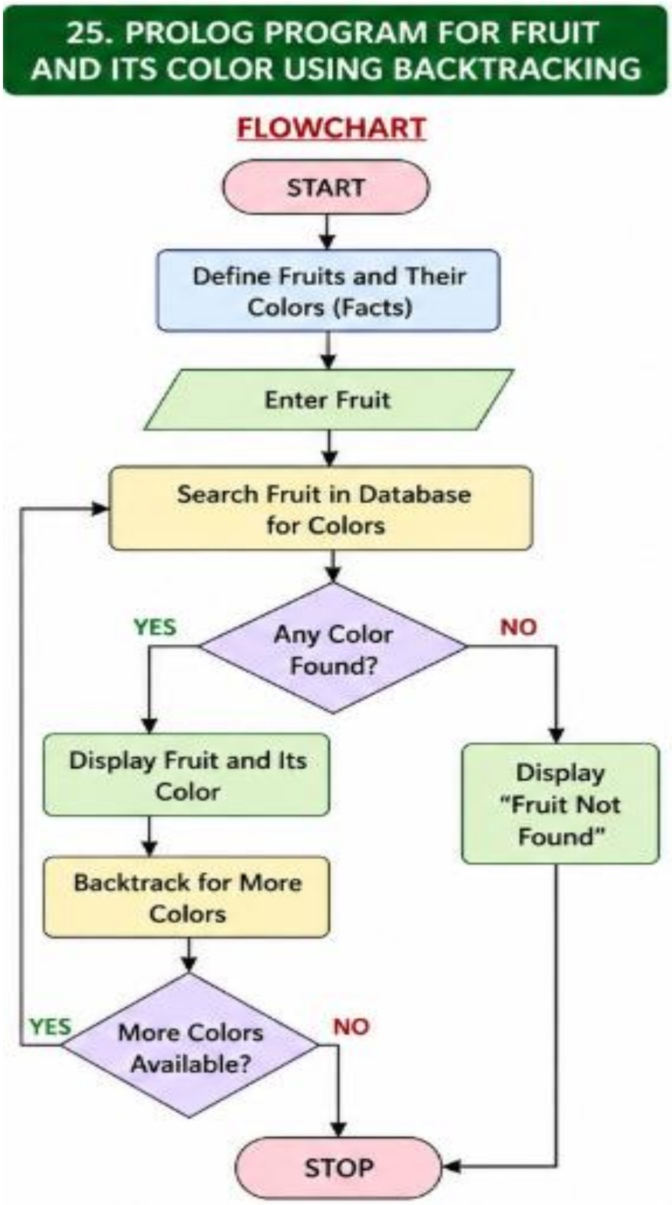
Objectives

- To understand backtracking in Prolog.
- To store fruit-color relationships.
- To retrieve multiple matching results.

Algorithm

1. Define fruits and their colors.
2. Select a fruit.
3. Search for its color.
4. Prolog backtracks to find additional matching facts.
5. Display the results.

Flowchart

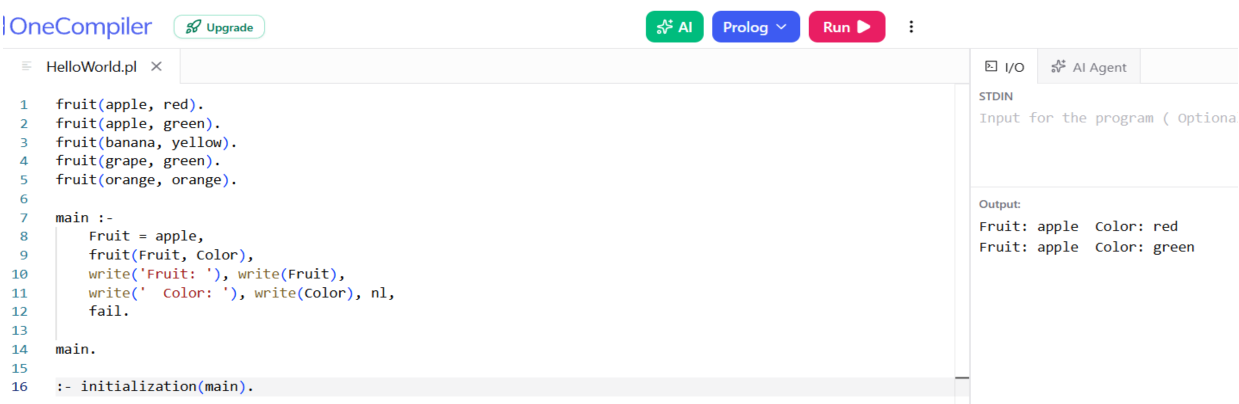


Code

```
fruit(apple, red).

fruit(apple, green).

fruit(banana, yellow).
```

```
fruit(grape, green).

fruit(orange, orange).

main :-

    Fruit = apple,

    fruit(Fruit, Color),

    write('Fruit: '),
write(Fruit),

    write('  Color: '),
write(Color), nl,

    fail.

main.

:- initialization(main).
```

Output



Result

Fruit: apple Color: red

Fruit: apple Color: green

Conclusion

Thus, the fruit and color relationship was successfully implemented using Prolog with backtracking.